TerraBit: Low-Bit Quantization of Geospatial Embedding Lakes 🌊

Isaac Corley
Independent Researcher
isaac.corley@proton.me

Abstract

Geospatial foundation models produce planet-scale *embedding lakes*—large, centralized stores of precomputed vectors analogous to data lakes—whose storage and retrieval costs dominate deployment budgets. Recent systems distribute global embedding fields at `int8` precision, but the quality–cost trade-off below 8 bits remains unexplored for remote sensing embeddings. We present **TerraBit**, a post-hoc quantization study spanning `float16` through 1-bit `binary` on 49,785,116 real 1024D embeddings (*183 GiB*), evaluated with label-free neighborhood-preservation metrics grounded in intrinsic-dimension analysis. `Int4` retains 91% kNN recall at $6.3\times$ compression; `binary` achieves $16.5\times$ compression with 65% recall, sufficient for coarse-stage candidate generation. We additionally benchmark TurboQuant [Zandieh et al., 2025], finding that its random orthogonal pre-rotation helps most in the ultra-low-bit regime: 2-bit recall rises from 0.54 to 0.71, while the advantage shrinks to near zero by 8 bits.

1 Introduction

Geospatial foundation models [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025] produce high-dimensional embeddings for retrieval, change detection, deduplication, and spatial indexing, forming *embedding lakes*—large, centralized vector stores analogous to data lakes. At planet scale, storage and memory bandwidth—not model accuracy—become the deployment bottleneck. A corpus of 50M embeddings at d=1024 in `float32` occupies *183 GiB*; halving per-vector cost saves nearly *100 GiB* with no retraining.

Several recent systems highlight this tension. AlphaEarth releases global embedding fields cast to `int8` via post-hoc calibration [Brown et al., 2025]. TESSERA applies quantization-aware training to a learned 128D bottleneck and distributes the result at `int8` [Feng et al., 2025]. ESD [Chen et al., 2026] takes a learned-compression approach, training a vector-quantized variational autoencoder (VQ-VAE)-style model with a Finite Scalar Quantization (FSQ) bottleneck [Mentzer et al., 2024] to compress Landsat+MODIS time series into `uint16` latent vectors, achieving $\sim$340$\times$ compression while improving downstream land-cover classification over raw reflectance (79.7% vs. 76.9%). These systems treat `int8`—or learned discrete codes—as the practical floor. **TerraBit** asks: *how far below `int8` can we push post-hoc quantization while preserving nearest-neighbor structure?*

We sweep seven bit-widths from `float16` to 1-bit `binary` on a real 49.8M-embedding corpus and measure neighborhood preservation with label-free metrics. Figure 2 shows graceful degradation in local-neighborhood preservation, and Table 1 quantifies recall@10 trade-offs for cosine and Euclidean distance. The frontier is smooth: `int4` retains 91% kNN recall at $6.3\times$ compression, while `binary` reaches $16.5\times$ with recall sufficient for shortlist retrieval [Wang et al., 2018].

Our contributions are: (1) an intrinsic-dimension-guided evaluation framework combining ID diagnostics [Levina and Bickel, 2004, Facco et al., 2017] with label-free retrieval fidelity metrics;

(2) a systematic post-hoc quantization study (`float16`, `fp8`, `int8`, `int4`, `int3`, `int2`, `binary`, and TurboQuant at four bit-widths) on 49.8M real geospatial embeddings; (3) a schema-preserving parallel Parquet transformation pipeline that processes the full corpus in 45 min at $18{,}359$ rows/s; (4) a retrieval benchmark showing $>1{,}300$ QPS on GPU with recall preserved across hardware; and (5) an empirical comparison showing that rotation-based quantization (TurboQuant) improves recall most at 2–3 bits, with diminishing gains as precision increases.

2 Related Work

Geospatial foundation models. Recent work trains large vision encoders on satellite imagery and releases pretrained embeddings as data products [Mendieta et al., 2023, Jakubik et al., 2024, Clay Foundation, 2025]. AlphaEarth [Brown et al., 2025] and TESSERA [Feng et al., 2025] both distribute global `int8` embedding fields, recognizing that storage cost matters as much as model quality at deployment scale.

Vector quantization and retrieval. Product quantization [Jégou et al., 2011], learned anisotropic quantization [Guo et al., 2020], and locality-sensitive hashing [Indyk and Motwani, 1998] compress embeddings for approximate nearest-neighbor search. FAISS [Johnson et al., 2021] implements these at billion scale. LLM.`int8`() [Dettmers et al., 2022] demonstrates that aggressive quantization preserves model quality in language; Matryoshka representations [Kusupati et al., 2022] show that embedding dimensions can be truncated with graceful degradation. TurboQuant [Zandieh et al., 2025] applies a random orthogonal pre-rotation before symmetric quantization to approach rate-distortion bounds in online vector quantization. For binary coding, prior work spans angular quantization for cosine retrieval [Gong et al., 2012], deep visual-semantic quantization with learned codebooks [Cao et al., 2017], and autoencoder-based near-lossless binarization in NLP [Tissier et al., 2019]. Orthogonally, learned codebook methods such as FSQ [Mentzer et al., 2024] replace VQ-VAE codebook lookup with per-dimension scalar rounding, enabling end-to-end discrete compression without codebook collapse—an approach recently applied to geospatial embeddings by ESD [Chen et al., 2026].

Positioning. Prior vector-compression work targets web-scale text/image embeddings or model-weight quantization; learned approaches like ESD [Chen et al., 2026] train task-specific autoencoders. **TerraBit** focuses on the complementary, less-studied regime of *post-hoc* quantization of geospatial embedding lakes, where spatial metadata must be preserved alongside vectors, no retraining is required, and label-free evaluation is the practical reality.

3 Method

Problem setup. Given a corpus $X = \{x_i\}_{i=1}^{N}$, $x_i \in \mathbb{R}^d$, stored in partitioned Parquet files with spatial/temporal metadata columns, we apply a quantizer $\hat{x}_i = \mathcal{Q}_m(x_i)$ under method m and evaluate each method by storage footprint, neighborhood preservation, and end-to-end throughput. Our corpus comprises $49{,}785{,}116$ embeddings (d=1024) produced by the Clay v1.5 foundation model [Clay Foundation, 2025], a ViT trained via masked autoencoding on multi-sensor satellite imagery.

Intrinsic-dimension diagnostics. Before compression, we estimate the intrinsic dimension (ID) of the embedding manifold using MLE [Levina and Bickel, 2004], TwoNN [Facco et al., 2017], and local PCA (LPCA) participation ratio across preprocessing variants (raw, centered, L2-normalized). ID estimates guide expectations for viable compression depth: if the data manifold has effective dimension $\hat{d} \ll d$, aggressive quantization should not collapse local neighborhood structure.

Quantization methods. We evaluate seven post-hoc methods spanning 16 bits to 1 bit: `float16` (IEEE half-precision truncation); `fp8` E4M3 (8-bit floating point with 4-bit exponent); `int`b for $b \in \{8, 4, 3, 2\}$ (per-dimension affine quantization with stored scale/zero-point vectors, followed by fixed-width bit packing for $b < 8$); and `binary` (sign-bit encoding, $\mathrm{sign}(x_i) \in \{0, 1\}^d$, packed to $d/8$ bytes, supporting Hamming-distance search [Wang et al., 2018]).

We additionally evaluate **TurboQuant** [Zandieh et al., 2025] at $b \in \{2, 3, 4, 8\}$ bits (`turbo`b). TurboQuant applies a fixed random orthogonal rotation $R \in \mathbb{R}^{d \times d}$ (sampled once from a Haar-

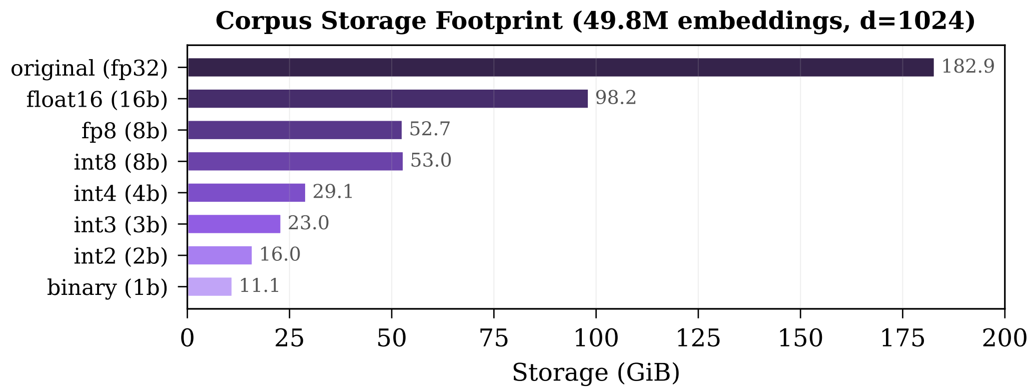


Figure 1: **Storage footprint per quantization method.** Original `float32` corpus occupies *183 GiB*. `Binary` reduces this to *11 GiB* (16.5×); `int4` to *29 GiB* (6.3×).

distributed ensemble via QR decomposition) before symmetric affine quantization: $\hat{x}_i = R^\top \mathcal{Q}_b(Rx_i)$. The rotation is intended to reduce inter-channel variance disparity, improving per-dimension range utilization.

Evaluation metrics. We report label-free, retrieval-oriented metrics: **kNN recall@**k ($k \in \{10, 25, 50\}$) under cosine and Euclidean distance, measuring the fraction of true k-nearest neighbors preserved after quantization; **rank correlation of cosine similarities**, the Spearman rank correlation of pairwise cosine similarities (original vs. quantized); and **reconstruction cosine similarity** and MSE (when dequantization is available).

Transformation pipeline. The pipeline processes Hive-partitioned Parquet files in parallel. For each file, it reads the embedding column, applies quantizers, replaces only that column, and writes output preserving directory structure, non-embedding columns, and schema. Quantization parameters (scale, zero-point, method, bit-width) are stored in Parquet footer metadata for downstream decompression.

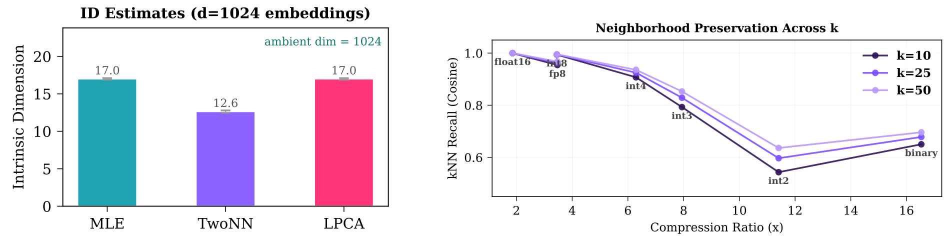
4 Experiments

Dataset. We evaluate on 49,785,116 embeddings (d=1024, `float32`) from a geospatial foundation model, stored across 1,021 Parquet files totaling *183 GiB*. Files are Hive-partitioned by level-2 geohash (511 cells), year, and month. Each record carries spatial metadata (geohash, bounding box, WKB geometry), temporal metadata, and provenance identifiers alongside the embedding vector.

Protocol. We evaluate along two axes: quality and systems efficiency. For quality, we compute label-free fidelity metrics on a 20,000-row reservoir sample, i.e., a uniform random sample maintained with reservoir sampling (reconstruction metrics are exact over all 49,785,116 rows). For systems, we run the full-corpus transformation over all 7 methods with 8 parallel workers, reporting wall-clock time and I/O throughput.

ID estimation on a 10,000-sample subset yields MLE $\hat{d}$ = 17.0, TwoNN $\hat{d}$ = 12.6, and LPCA $\hat{d}$ = 17.0 (Figure 2a), confirming substantial structural redundancy relative to ambient dimension d=1024.

Table 1 summarizes quality across all methods. `Int8` yields near-lossless retrieval (recall@10 = 0.99) at 3.5× compression, matching the precision used by AlphaEarth and TESSERA [Brown et al., 2025, Feng et al., 2025]. `Int4` provides a strong balanced operating point: 6.3× compression with 91% kNN recall. `Binary` achieves the highest compression (16.5×) with recall@10 of 0.65—suitable as a coarse first-pass filter before re-ranking with higher-fidelity representations [Wang et al., 2018, Jégou et al., 2011].



(a) ID estimates. MLE, TwoNN, and LPCA cluster at $\hat{d} \approx$ 13–17, far below d=1024.

(b) kNN recall (cosine) across $k \in \{10, 25, 50\}$. Recall declines smoothly as bit-width decreases.

Figure 2: **Intrinsic dimension and neighborhood preservation.** Low intrinsic dimension motivates aggressive quantization, and recall degrades gracefully across operating points.

Table 1: **Quality–compression results on** $49{,}785{,}116$ **embeddings.** Reconstruction cosine is exact over all rows; kNN recall and cosine correlation are sampled from a 20,000-row reservoir. The best compression ratio is shown in **bold**.

Method	Ratio ($\times$)	Recon. Cos $\uparrow$	C@10 $\uparrow$	E@10 $\uparrow$	Cos Corr $\uparrow$
`float16`	1.86	1.00	1.00	1.00	1.00
`fp8`	3.47	1.00	0.95	0.95	1.00
`int8`	3.45	1.00	0.99	0.99	1.00
`int4`	6.29	1.00	0.91	0.91	1.00
`int3`	7.94	0.98	0.79	0.79	0.99
`int2`	11.41	0.91	0.54	0.53	0.95
`binary`	**16.53**	0.49	0.65	0.66	0.79

TurboQuant analysis. Table 2 and Figure 3 compare TurboQuant [Zandieh et al., 2025] against standard `int`b at matched bit-widths. The headline finding is the opposite of our initial expectation: turbo methods improve recall at every tested bit-width, but the gain shrinks rapidly as precision increases.

Largest gains at the lowest bit-widths. At 2 bits, `turbo2` reaches recall@10 of 0.71 versus 0.54 for `int2`, a gain of 16 points at essentially identical compression. At 3 bits, `turbo3` improves recall from 0.79 to 0.88. The gap narrows at 4 bits (0.95 vs. 0.91) and is nearly zero at 8 bits (1.00 vs. 0.99).

Why the gain diminishes with precision. TurboQuant's random orthogonal rotation redistributes variance across channels before scalar quantization. That equalization matters most when the quantizer is extremely coarse: at 2–3 bits, better range utilization produces a meaningful recall gain. Once bit-width is high enough that plain per-dimension affine quantization is already near-lossless, the extra rotation has little room to help. This pattern fits the geometry of geospatial ViT embeddings: the manifold is low-dimensional enough that 8-bit quantization already preserves neighborhood structure almost perfectly, so rotation mainly benefits the most constrained operating points.

Takeaway. For geospatial embedding lakes, TurboQuant is most useful in the ultra-low-bit regime. If the deployment target is 2–3 bits, rotation offers a measurable quality gain; if 8-bit storage is acceptable, plain affine quantization is simpler and effectively equivalent.

Improving binarization without retraining. To test whether binary quality can be improved in the strict post-hoc regime, we evaluate three lightweight variants: per-dimension median thresholding (`binary_med`), z-score thresholding (`binary_zscore`), and an ITQ-style PCA-whitened rotation (`binary_itq`). We run `binary_med` and `binary_zscore` on the full corpus; `binary_itq` is evaluated on the subsample only because its learned full-dimensional transform introduces large per-file metadata overhead in the current Parquet format. On a 20,000-embedding subsample, `binary_med` appears promising, improving cosine recall@10 from 0.637 to 0.659 at the same 32$\times$ code-only compression. However, Figure 4 shows that this gain does *not* survive at corpus scale: on the full corpus, `binary_med` drops to 0.40 recall@10 and underperforms plain sign coding.

Table 2: **TurboQuant [Zandieh et al., 2025] vs. standard** `int`b **on** 49,785,116 **embeddings.** Compression ratios are near-identical; TurboQuant improves recall most at low bit-widths. The advantage shrinks steadily as precision increases.

Method	Ratio (×)	Recon. Cos ↑	C@10 ↑	E@10 ↑	Cos Corr ↑
`int2`	11.41	0.91	0.54	0.53	0.95
`turbo2`	11.44	0.91	0.71	0.69	0.99
`int3`	7.94	0.98	0.79	0.79	0.99
`turbo3`	7.96	0.98	0.88	0.88	1.00
`int4`	6.29	1.00	0.91	0.91	1.00
`turbo4`	6.10	1.00	0.95	0.95	1.00
`int8`	3.45	1.00	0.99	0.99	1.00
`turbo8`	3.45	1.00	1.00	1.00	1.00

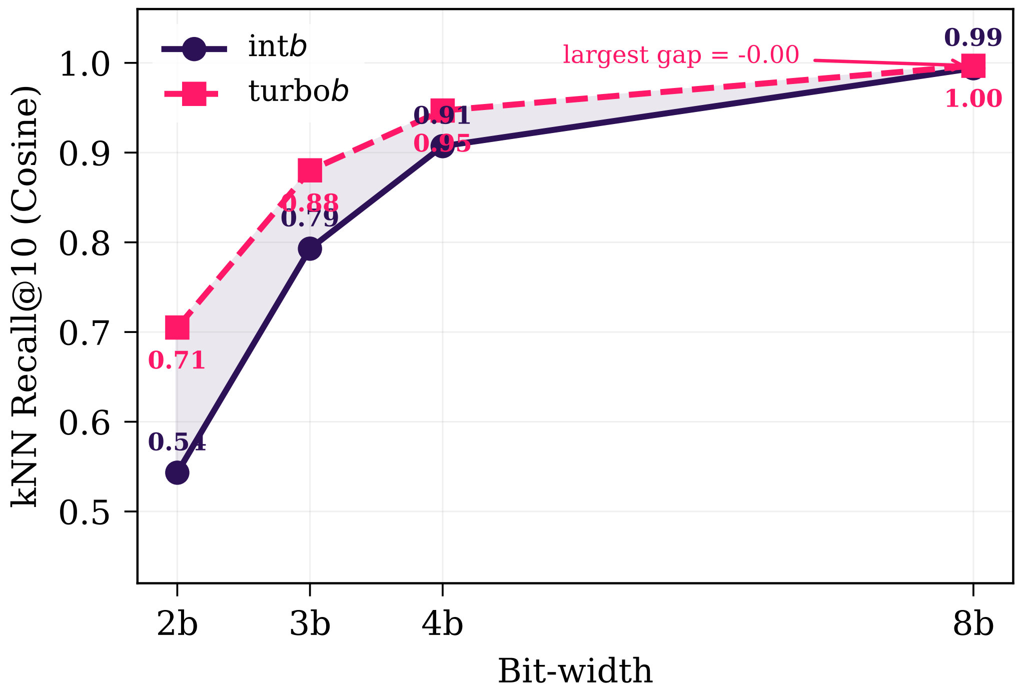


Figure 3: **Standard** `int`b **vs. TurboQuant recall@10 across bit-widths.** Cosine recall@10 across matched bit-widths. Shaded region marks the recall gap. TurboQuant improves recall at every matched bit-width, with the largest gain at 2 bits and a near-zero gap by 8 bits. Rotation helps most when quantization is extremely coarse, then becomes negligible once affine quantization is already near-lossless.

This failure mode is informative. Both variants estimate thresholds independently within each Parquet partition, so the same semantic region of embedding space is binarized with different decision boundaries across geohash/year shards. That per-file calibration drift is mostly invisible on a local subsample but becomes destructive when nearest neighbors span many partitions. The result also reinforces a broader point from angular-hashing work [Gong et al., 2012]: preserving rank neighborhoods in binary space depends on consistent angular partitions, not merely on improving local reconstruction statistics. Indeed, `binary_zscore` attains very high reconstruction cosine (0.925) yet very poor recall@10 (0.310), showing that reconstruction fidelity is a poor proxy for binary retrieval quality.

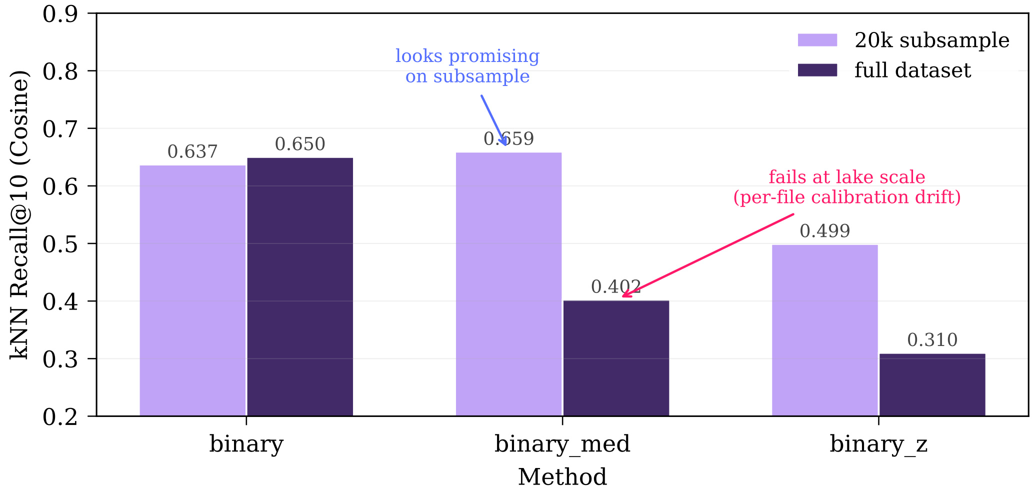


Figure 4: **Binary variants on subsample vs. full corpus.** `binary_med` improves over plain sign coding on a $20{,}000$-embedding subsample, but the advantage reverses on the full corpus. This exposes a scale-sensitive failure mode: per-file threshold calibration does not transfer across partitions. Plain `binary` remains the strongest lake-scale binary baseline.

Practically, the best-performing post-hoc binary method in our lake-scale setting is still the simplest one: global sign coding with no per-file calibration. This suggests that future binary improvements should use *global* calibration or shared transforms rather than independent per-partition thresholds.

Storage and throughput. Figure 1 shows the per-method storage footprint. `Binary` reduces the *183 GiB* corpus to *11 GiB*; `int4` to *29 GiB*. The full pipeline (all 7 methods, 8 workers) processes the entire corpus in $45.2\,\mathrm{min}$ at 18,359 rows/s and 107 $\mathrm{MiB/s}$ output throughput.

Discussion. Four findings emerge. First, ID estimates of 13–20 (vs. ambient $d{=}1024$) explain why aggressive quantization does not immediately collapse neighborhood structure: most per-dimension variance is redundant for local geometry. Second, the quality decline from `int8` to `int2` is gradual (Figure 2b), enabling practitioners to select operating points based on their recall requirements versus storage budget. Third, `binary` quantization is not quality-preserving in a strict sense, but its $16.5\times$ compression and compatibility with Hamming-distance search make it attractive for first-pass candidate generation in multi-stage retrieval pipelines [Wang et al., 2018]. Fourth, post-hoc binary refinements that look promising on small subsamples can fail at lake scale (Figure 4); robustness across partitions matters more than local reconstruction quality, so future work should prefer shared/global calibration over per-file binary transforms.

Retrieval benchmark. To validate that quantization-induced compression translates to practical search speedups, we benchmark brute-force kNN retrieval on a 1M-vector subset ($1{,}000$ queries, $k{=}50$) using FAISS [Johnson et al., 2021] on CPU and PyTorch `cdist` on a single NVIDIA RTX 3090 GPU. Each quantized corpus is dequantized to `float32` before indexing, isolating the effect of quantization noise from format-specific index optimizations; binary embeddings use Hamming distance directly via packed `uint8` search.

Table 3 and Figure 5 summarize throughput and recall. GPU search achieves $1{,}140$–$1{,}371$ QPS for dequantized methods—a $22\times$–$28\times$ speedup over FAISS CPU (48–60 QPS)—with matched recall, confirming that quantization quality is hardware-independent. Binary Hamming search runs at 270 QPS on CPU and 134 QPS on GPU; the lower GPU throughput reflects the overhead of bit-unpacking binary vectors into float tensors for `cdist`, a limitation of the brute-force approach that purpose-built Hamming kernels would eliminate.

Limitations. We use label-free neighborhood-preservation proxies rather than task-specific metrics (e.g., land-cover classification accuracy). kNN recall and cosine correlation are computed on a

Table 3: **Brute-force search benchmark** on 1M embeddings (1K queries, k=10). QPS = queries per second. GPU is a single RTX 3090. Recall is identical across backends since quantization quality is hardware-independent. **Bold**: best throughput per row.

Method	CPU QPS	GPU QPS	Speedup	R@10
`float16`	60	**1322**	22.2×	1.00
`fp8`	49	**1238**	25.5×	0.95
`int8`	49	**1140**	23.5×	0.99
`int4`	48	**1371**	28.3×	0.90
`int3`	48	**1224**	25.3×	0.81
`int2`	48	**1157**	24.4×	0.61
`binary`	**270**	134	0.5×	1.00*

*Recall measured against binary GT (Hamming distance).

reservoir sample, not the full corpus. Our retrieval benchmark uses brute-force search; approximate nearest-neighbor indices such as inverted file (IVF) and hierarchical navigable small worlds (HNSW) may alter the quality–throughput frontier at larger scale [Johnson et al., 2021]. Future work should add labeled downstream evaluation (e.g., land-cover classification accuracy as in Chen et al. [2026]), investigate ANN index interactions with low-bit quantization, and explore whether stacking post-hoc quantization on top of learned discrete representations [Mentzer et al., 2024, Chen et al., 2026] can compound compression gains.

5 Conclusion

TerraBit provides a systematic compression study for geospatial embedding lakes. On 49.8M real embeddings, low-bit quantization yields a smooth quality–cost frontier: `int4`/`int8` are strong balanced choices; `binary` is viable for coarse retrieval stages. GPU-accelerated brute-force search

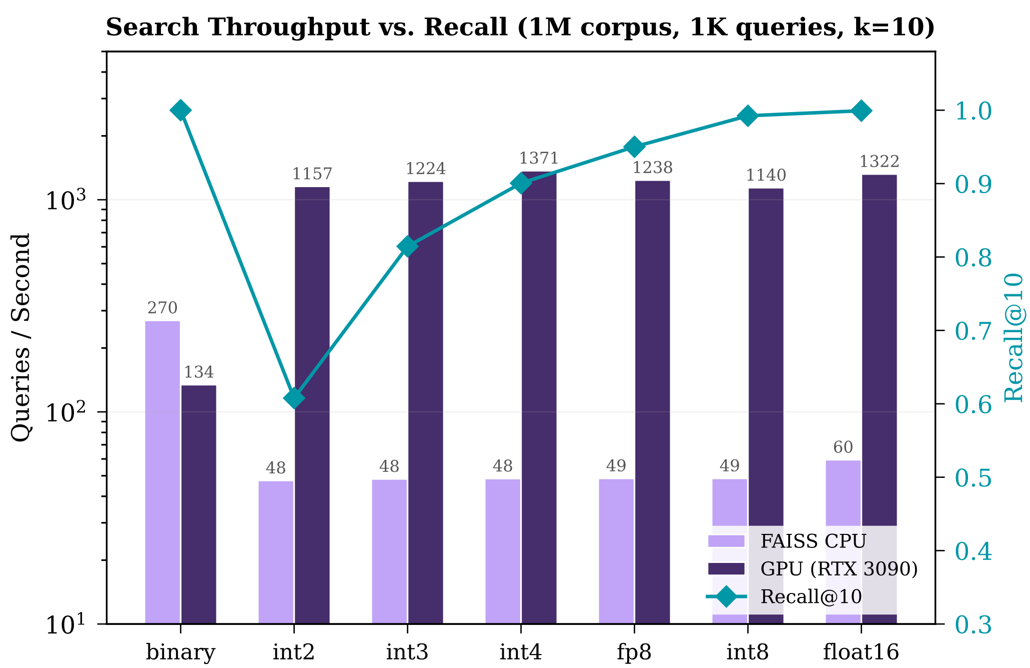


Figure 5: **Search throughput vs. recall** on 1M embeddings. GPU brute-force delivers 1,140–1,371 QPS across dequantized methods. Binary Hamming search is fastest on CPU (270 QPS) due to native bitwise ops. Recall (purple diamonds, right axis) tracks quantization fidelity independently of hardware.

over dequantized embeddings exceeds 1,300 QPS on a single RTX 3090, demonstrating that quantized embedding lakes are practical for real-time retrieval at scale.

References

Christopher F. Brown, Michal R. Kazmierski, Valerie J. Pasquarella, William J. Rucklidge, Masha Samsikova, Chenhui Zhang, Evan Shelhamer, Estefania Lahera, Olivia Wiles, Simon Ilyushchenko, Noel Gorelick, Lihui Lydia Zhang, Sophia Alj, Emily Schechter, Sean Askay, Oliver Guinan, Rebecca Moore, Alexis Boukouvalas, and Pushmeet Kohli. AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. *arXiv preprint arXiv:2507.22291*, 2025.

Yue Cao, Mingsheng Long, Jianmin Wang, and Shichen Liu. Deep visual-semantic quantization for efficient image retrieval. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1328–1337, 2017.

Shuang Chen, Jie Wang, Shuai Yuan, Jiayang Li, Yu Xia, Yuanhong Liao, Junbo Wei, Jincheng Yuan, Xiaoqing Xu, Xiaolin Zhu, Peng Zhu, Hongsheng Zhang, Yuyu Zhou, Haohuan Fu, Huabing Huang, Bin Chen, Fan Dai, and Peng Gong. Democratizing planetary-scale analysis: An ultra-lightweight earth embedding database for accurate and flexible global land monitoring. *arXiv preprint arXiv:2601.11183*, 2026.

Clay Foundation. Clay foundation model v1.5. `https://clay-foundation.github.io/model/`, 2025. Apache-2.0 license.

Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, 2022.

Elena Facco, Maria d'Errico, Alex Rodriguez, and Alessandro Laio. Estimating the intrinsic dimension of datasets by a minimal neighborhood information. *Scientific Reports*, 7(1):12140, 2017.

Zhengpeng Feng, Clement Atzberger, Sadiq Jaffer, Jovana Knezevic, Silja Sormunen, Robin Young, Madeline C. Lisaius, Markus Immitzer, Toby Jackson, James Ball, David A. Coomes, Anil Madhavapeddy, Andrew Blake, and Srinivasan Keshav. TESSERA: Temporal embeddings of surface spectra for earth representation and analysis. *arXiv preprint arXiv:2506.20380*, 2025.

Yunchao Gong, Sanjiv Kumar, Vishal Verma, and Svetlana Lazebnik. Angular quantization-based binary codes for fast similarity search. In *Advances in Neural Information Processing Systems*, 2012.

Ruiqi Guo, Philip Sun, Erik Lindgren, Quan Geng, David Simcha, Felix Chern, and Sanjiv Kumar. Accelerating large-scale inference with anisotropic vector quantization. In *International Conference on Machine Learning*, pages 3887–3896, 2020.

Piotr Indyk and Rajeev Motwani. Approximate nearest neighbors: Towards removing the curse of dimensionality. In *Proceedings of the Thirtieth Annual ACM Symposium on Theory of Computing*, pages 604–613, 1998.

Johannes Jakubik, Sujit Roy, C. E. Phillips, Paolo Fraccaro, Denys Godwin, Birgit Zadrozny, Daniela Szwarcman, Carlos Gomes, Gabby Nyirjesy, Blair Edwards, Manil Maskey, et al. Foundation models for generalist geospatial artificial intelligence. *arXiv preprint arXiv:2310.18660*, 2024.

Hervé Jégou, Matthijs Douze, and Cordelia Schmid. Product quantization for nearest neighbor search. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(1):117–128, 2011.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2021.

Aditya Kusupati, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sber, Prateek Jain, Sham Kakade, and Ali Farhadi. Matryoshka representation learning. In *Advances in Neural Information Processing Systems*, 2022.

Elizaveta Levina and Peter J. Bickel. Maximum likelihood estimation of intrinsic dimension. *Advances in Neural Information Processing Systems*, 17, 2004.

Matias Mendieta, Boran Han, Xingjian Shi, Yi Zhu, and Chen Chen. Towards geospatial foundation models via continual pretraining. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16806–16816, 2023.

Fabian Mentzer, David Minnen, Eirikur Agustsson, and Michael Tschannen. Finite scalar quantization: VQ-VAE made simple. In *International Conference on Learning Representations*, 2024.

Julien Tissier, Christophe Gravier, and Amaury Habrard. Near-lossless binarization of word embeddings. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(1):7104–7111, 2019.

Jingdong Wang, Ting Zhang, Jingkuan Song, Nicu Sebe, and Heng Tao Shen. A survey on learning to hash. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(4):769–790, 2018.

Amir Zandieh, Majid Daliri, Majid Hadian, and Vahab Mirrokni. TurboQuant: Online vector quantization with near-optimal distortion rate. *arXiv preprint arXiv:2504.19874*, 2025.